CANCER DATA ANALYSIS

REPORT

Data Analysis Project

Author: Shravan Aldar

Tools Used:

Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy, Statsmodels, Scikit-learn)

1. Abstract

This project analyzes a global dataset of **50,000 cancer patient records (2015–2024)** across 10 countries and 8 cancer types. Using Python, it explores demographics, risk factors, cancer stages, treatment costs and survival. The analysis combined descriptive statistics, visualization, correlation and regression analysis, hypothesis tests (Pearson, Spearman, Kruskal-Wallis, OLS interaction) and Random Forest models.

Smoking and genetic risk are the strongest predictors of the severity score, followed by alcohol use, air pollution and obesity. Treatment cost and survival years showed no meaningful relationship with cancer stage, treatment cost, demographics or risk factors. Early-stage diagnosis (Stage 0 and I) was consistent across cancer types at about 38–41%.

2. Introduction

Cancer is a major global health burden. Understanding which risk factors drive severity, how early cancers are diagnosed and how costs vary across demographics can guide prevention and policy. This project applies the full data analysis workflow of cleaning, exploring, visualizing, testing and modeling to a global cancer patient dataset.

3. Objectives

1. Perform descriptive analysis of the dataset.
2. Determine the relationship between risk factors and cancer severity.
3. Analyze the proportion of early-stage diagnoses by cancer type.
4. Identify key predictors of cancer severity and survival years.

5. Explore the economic burden of treatment across countries and demographics.
6. Test whether higher treatment cost is associated with longer survival.
7. Evaluate whether higher cancer stage leads to greater cost and reduced survival.
8. Examine whether genetic risk amplifies the effect of smoking on severity.

4. Dataset Description

- **File:** global_cancer_patients_2015_2024.csv
- **Size:** 50,000 rows × 15 columns (8 float, 2 integer, 5 text)
- **Data quality:** no missing values and no duplicate rows

Column	Description
Patient_ID	Unique patient identifier
Age	Patient age (20–89)
Gender	Male / Female / Other
Country_Region	10 countries
Year	Year of record (2015–2024)
Genetic_Risk, Air_Pollution, Alcohol_Use, Smoking, Obesity_Level	Risk-factor scores on a 0–10 scale
Cancer_Type	8 types (Lung, Breast, Colon, Liver, Skin, Leukemia, Cervical, Prostate)
Cancer_Stage	Stage 0, I, II, III, IV

Column	Description
Treatment_Cost_USD	Cost of treatment ($5,000–$100,000)
Survival_Years	Years survived (0–10)
Target_Severity_Score	Severity score (0.9–9.16)

5. Methodology

1. **Data loading and inspection:** info(), describe() and duplicate checks with Pandas.
2. **Univariate analysis:** histograms, KDE plots, bar charts and pie charts for each key variable.
3. **Relationship analysis:** linear regression (linregress) of each risk factor against severity.
4. **Correlation analysis:** Pearson and Spearman correlations with severity and survival.
5. **Feature engineering:** created Age_Group bins (0–30, 31–45, 46–60, 61–75, 76+).
6. **Modeling:** Random Forest regressors (label-encoded categories, 80/20 train-test split) for severity and survival, with GridSearchCV tuning for survival.
7. **Hypothesis testing:** Pearson and Spearman tests ($\alpha = 0.05$), Shapiro-Wilk normality test, Kruskal-Wallis test, and OLS regression with an interaction term.

6. Analysis and Findings

6.1 Descriptive analysis

Variable	Finding
Age	Mean 54.4, median 54, range 20–89. Distribution is nearly uniform.
Gender	Male 16,796 · Female 16,709 · Other 16,495 (balanced)
Country	4,864–5,092 patients per country (about 10% each)
Cancer type	Colon 6,376 · Prostate 6,308 · Leukemia 6,266 · Liver 6,249 · Skin 6,231 · Cervical 6,222 · Breast 6,189 · Lung 6,159
Cancer stage	Stage II 10,124 · I 10,046 · III 10,008 · IV 9,933 · 0 9,889
Treatment cost	Mean $52,467, median $52,474, std $27,363, range $5,000–$100,000, evenly distributed
Survival years	Mean 5.01, range 0–10, roughly uniform
Severity score	Mean 4.95, std 1.20
Risk factors	All five have a mean of about 5.0 and std of about 2.89 on a 0–10 scale

The dataset is well balanced across gender, country, cancer type and stage.

6.2 Risk factors vs. severity (linear regression)

Risk Factor	R^2	Slope
Smoking	0.23	0.20
Genetic Risk	0.23	0.20
Air Pollution	0.13	0.15
Alcohol Use	0.13	0.15
Obesity Level	0.06	0.10

All five factors show a positive relationship with severity. Smoking and genetic risk have the strongest individual effect, and obesity the weakest.

6.3 Early-stage diagnosis by cancer type (Stage 0 + Stage I)

Cancer Type	Early-stage %
Liver	40.61
Colon	40.42
Skin	40.41
Prostate	40.19
Cervical	39.86
Leukemia	39.53
Breast	39.47
Lung	38.43

Overall, about 40% of patients are diagnosed early. Lung cancer has the lowest rate and liver the highest, but the full gap is only about 2 percentage points.

6.4 Key predictors

Correlation with severity (Pearson):

Factor	Correlation
Smoking	0.484
Genetic Risk	0.479
Air Pollution	0.367
Alcohol Use	0.363
Obesity Level	0.251
Age	−0.001

Spearman results are nearly identical. Correlation with survival years is about 0 ($|r| < 0.01$) for every variable.

Random Forest for severity (Train $R^2 = 0.97$, Test $R^2 = 0.78$): approximate feature importance was Smoking 0.26, Genetic Risk 0.25, Alcohol Use 0.16, Air Pollution 0.16, Obesity 0.085. Age, country, year, cancer type, stage and gender were each below 0.03.

Random Forest for survival (tuned with GridSearchCV; Train $R^2 = 0.009$, Test $R^2 \approx 0.00$): the model cannot predict survival from the available features.

6.5 Economic burden across countries and demographics

Country	Avg. cost (USD)
China	52,899
USA	52,879
Germany	52,769
Australia	52,622
Canada	52,584
Brazil	52,541
Russia	52,319
India	52,285
UK	52,200
Pakistan	51,568

- **Country:** the gap between the highest and lowest country is only about $1,300 (about 2.5%).
- **Gender:** Female $52,091, Male $52,564, Other $52,749. The difference is minimal.
- **Age group:** averages range from $52,128 (31–45) to $52,783 (76+). There is no clear rise with age.
- **Heatmap (country × age group):** cell values stay within about $49,700–$54,900, so cost is nearly uniform across demographics.

6.6 Treatment cost vs. survival

- H_0: no correlation between treatment cost and survival years.

- Pearson r = −0.0004 (p = 0.924); Spearman ρ = −0.0004 (p = 0.921).
- **Result:** fail to reject H_0. Higher treatment cost is not associated with longer survival.

6.7 Cancer stage vs. cost and survival

Stage	Avg. Cost (USD)	Avg. Survival (years)
Stage 0	52,573	5.02
Stage I	52,674	5.01
Stage II	52,083	5.00
Stage III	52,708	5.04
Stage IV	52,302	4.97

The Shapiro-Wilk test showed all groups are non-normal ($p < 0.05$), so the **Kruskal-Wallis test** was used.

- Cost across stages: p = 0.425, so no significant difference.
- Survival across stages: p = 0.603, so no significant difference.

6.8 Interaction: genetic risk × smoking

OLS model: Target_Severity_Score ~ Genetic_Risk * Smoking

- Interaction coefficient = −0.000228, 95% CI [−0.00115, 0.00069], p = 0.628.
- **Result:** fail to reject H_0. Genetic risk does not amplify the effect of smoking. The two act independently on severity.

7. Key Insights

1. **Smoking and genetic risk** are the strongest drivers of severity, followed by alcohol, air pollution and obesity.
2. **Smoking, alcohol, air pollution and obesity are modifiable**, which makes them targets for prevention.
3. **About 40% of cancers are diagnosed early**, with lung cancer lowest at 38.4%.
4. **Treatment cost is uniform** across countries, genders and age groups.
5. **Survival years show no relationship** with cost, stage, risk factors or demographics.
6. **Risk factors act independently**, with no genetic × smoking interaction.

8. Recommendations

- Prioritize **smoking-cessation and awareness programs**, since smoking is the top modifiable driver.
- Reduce exposure to **air pollution and alcohol** through public health policy.
- Strengthen **lung cancer screening** to improve early detection.
- Investigate why survival is unrelated to stage and cost, since real-world data usually shows a link.
- Add clinical variables (treatment type, comorbidities, follow-up dates) to enable better survival modeling.

9. Conclusion

The analysis shows that smoking and genetic risk are the dominant factors behind cancer severity, while the dataset shows no meaningful link between survival and any other variable. Severity is predictable (test R^2 of about 0.78), but survival is

not (R^2 of about 0). Costs are almost identical across countries, genders and age groups.

10. Limitations and Future Scope

Limitations:

- Most variables are evenly (uniformly) distributed and survival has no relationship with any variable, which suggests the data may be simulated rather than real-world.
- The severity Random Forest overfits (train R^2 0.97 vs. test R^2 0.78).
- Label encoding was used for categorical variables.
- The dataset has no clinical details such as treatment type, comorbidities or follow-up.

Future scope:

- Use cross-validation and feature selection to reduce overfitting.
- Try gradient boosting (XGBoost) and one-hot encoding.
- Build survival analysis models (Kaplan-Meier, Cox regression) on real clinical data.